		distance between the towers = 11 x 10 x 0.000275 = 3.0 m
19	A	0 because no change in electric potential
20	C	Option C: $E = -\frac{dV}{dr}$ hence potential gradient is numerically (magnitude) equal to the electric field strength at that point. The statement is true. Option A: For a positive charge, the potential decreases at a decreasing rate away from the positive charge. Option B: The negative of the potential gradient is the electric field strength at that point. Thus option B is wrong as potential gradient represents a vector. Option D: Electric field strength at a point is the electric force acting on a charge per unit positive charge placed at that point.
21	A	At I = 0.1 A, V across resistor (1.0V) and V across thermistor (2.0) adds up to be the e.m.f of the battery